

$$9) f(x) = \frac{x^2 + 4x + 3}{x + 3}$$

$$10) f(x) = \frac{x}{x^2 - 4x}$$

$$11) f(x) = \begin{cases} x + 4, & x \leq -2 \\ -2x - 11, & x > -2 \end{cases}$$

$$12) f(x) = \frac{x + 7}{x^2 + 3x}$$

Find the intervals on which each function is continuous.

$$13) f(x) = \begin{cases} x, & x \neq 4 \\ 2, & x = 4 \end{cases}$$

$$14) f(x) = \begin{cases} -2, & x < 3 \\ -2x + 6, & x \geq 3 \end{cases}$$

$$15) f(x) = \frac{x - 1}{x^2 - 4x + 3}$$

$$16) f(x) = \frac{x^2}{2} + 4x + 10$$

$$17) f(x) = -x^2 - 4x + 2$$

$$18) f(x) = -\frac{x - 2}{x^2 - 3x + 2}$$

$$19) f(x) = -\frac{x - 1}{x^2 - x}$$

$$20) f(x) = \frac{x}{x^2 - 6x + 9}$$

Critical thinking questions:

21) Write a function that has an infinite discontinuity at $x = 100$.

22) Write a function that is continuous over $(-\infty, 0)$, $(0, 1)$, and $(1, \infty)$ and discontinuous everywhere else.